

9(a)	<i>any six from</i> 1 different substrates have different numbers of, hydrogens / C-H bonds ; 2 lipids have (relatively) more, hydrogens / C-H bonds (than carbohydrates or proteins) ; 3 hydrogens / C-H bonds, located in fatty acid (tails of lipids) ; 4 breakdown / oxidation, of substrate provides hydrogen (atoms) ; 5 for reduction of, NAD / FAD ; 6 (reduced, NAD / FAD) provides / releases, hydrogen to ETC ; 7 hydrogen (dissociates) into protons and electrons ; 8 <i>ref.</i> energy used to set up proton gradient ; 9 chemiosmosis / oxidative phosphorylation / AW ; 10 (so) more, ATP / energy, from lipids per unit mass (than, carbohydrates / proteins) or lipids, more energy dense / have higher (relative) energy value ;	6
9(b)	<i>RQ</i> 1 (ratio of) carbon dioxide given out divided by oxygen taken in ; 2 <i>ref.</i> volume / moles ; 3 R amount per unit time ; <i>any eight from investigation</i> 4 use respirometer ; 5 seeds placed on, mesh / gauze ; 6 KOH / NaOH / sodalime, to absorb carbon dioxide ; 7 manometer / capillary tube / syringe ; 8 movement of fluid (in manometer / capillary tube / syringe) = uptake of oxygen ; 9 keep, temperature / air pressure, constant ; 10 measure oxygen uptake after certain time ; 11 repeat without KOH / NaOH / sodalime ; 12 difference in manometer readings due to carbon dioxide given out ;	9